

Push vs. Just-In-Time vs. Just-In-Sequence: A Discrete-Event Simulation Comparison of Production System Trade-offs in Automotive Mixed-Model Assembly

Research Project for Industrial Engineering

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Abstract

Background: Automotive manufacturers face mounting pressure to reduce inventory waste and improve supply chain responsiveness while maintaining throughput in mixed-model assembly environments. The choice between Push, Just-In-Time (JIT), and Just-In-Sequence (JIS) production systems represents a fundamental trade-off with significant operational and financial consequences for original equipment manufacturers (OEMs).

Objective: This study aimed to develop and validate a unified discrete-event simulation (DES) framework enabling direct, quantitative comparison of Push, JIT, and JIS production systems under identical, controlled conditions.

Methodology: Three AnyLogic Process Modeling Library simulation models were constructed using parameters sourced exclusively from 11 peer-reviewed publications. A total of 31 replications were conducted across three scenarios (10 Push, 10 JIT, 11 JIS Disrupted) over a simulated 480-minute shift. Five KPIs were measured: throughput, WIP inventory, lead time, station utilisation, and sequence accuracy.

Results: JIT reduced WIP inventory by 73.9% (37.1 → 9.7 cars) and lead time by 41.5% (115.4 → 67.5 minutes) relative to Push, while throughput variability increased 11.8-fold (CV: 0.054 → 0.638). JIS produced a mean of 30.8 sequence errors per shift, resulting in 462 minutes of line-stoppage dead-time per 480-minute shift and a mean throughput of 36.3 ± 14.1 cars — a 54.5% reduction relative to JIT.

Conclusion: The Push–JIT–JIS trade-off curve established by this study provides OEMs with quantified, evidence-based guidance for production system selection. JIT is recommended for facilities with high supplier reliability; JIS requires active resequencing buffers before adoption. This study contributes the first unified, reproducible DES comparison of all three systems under identical experimental conditions, calibrated exclusively from peer-reviewed literature.

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1. INTRODUCTION

Every 4.4 seconds one vehicle rolls off a global automotive assembly line yet the production philosophy governing how that vehicle is built can mean the difference between a lean, responsive factory and one drowning in excess inventory and extended lead times. In 2023, the global automotive sector produced over 93 million vehicles while facing mounting pressure to reduce waste, shorten delivery windows, and accommodate unprecedented product variety (Andjelković et al., 2025). At the core of this challenge lies a fundamental strategic choice: which production replenishment system — Push, Just-In-Time (JIT) or maybe Just-In-Sequence (JIS) best serves a modern mixed-model automotive assembly operation?

Background

Background

Miniature literature review — what has been done and where it ends.

The traditional **Push production system**, grounded in Material Requirements Planning schedules output from demand forecasts rather than actual downstream signals. While operationally predictable, this approach consistently generates excessive Work-In-Progress inventory at every buffer stage, conceals process inefficiencies, and inflates customer lead times. The 1973 oil crisis exposed these inefficiencies at scale compelling Japanese manufacturers to seek fundamentally leaner alternatives.

In response, **Just-In-Time (JIT)** emerged from Toyota's Production System, codified by Monden (1983). Driven by Kanban pull signals, JIT delivers materials precisely when needed. Huson and Nanda (1995) documented a 24% improvement in inventory turnover following JIT adoption. Pawlewski et al. (2012), using FlexSim discrete-event simulation, demonstrated that JIT reduced container requirements from 20 (Push baseline) to an optimal 11 — a 45% inventory reduction. However, JIT's lean design introduces fragility: with minimal buffer stock, any supplier delay can starve the production line and collapse throughput.

As manufacturers shifted toward mass customisation, JIT revealed structural limitations. **Just-In-Sequence (JIS)** was developed to address product variety complexity, ensuring parts

arrive not only on time but in the precise build sequence matching each vehicle — eliminating warehouse staging entirely Real German OEM data confirms JIS seat delivery at 55–65 second takt intervals with batches of 15 seat sets Yet sequence errors at approximately 5% frequency trigger 15-minute line stoppages per event, converting precision sequencing into a potential throughput liability

Discrete-Event Simulation (DES) has established itself as the standard methodology for evaluating production systems without disrupting live operations. Wang and Chen (2016) modelled a Chinese truck assembly line in AnyLogic establishing station time benchmarks directly applicable to this study. Sand et al. (2023) validated stochastic supplier lead time distributions across 100-replication DES experiments. Mahin et al. (2024) established an AnyLogic throughput benchmark of approximately 14 vehicles per hour, providing an independent validation target.

Research Problem

Research Problem

Defining the specific gap

Despite extensive individual investigations, a critical gap persists: **no single validated simulation framework has simultaneously compared Push, JIT, and JIS under identical, literature-calibrated conditions**. Studies examining each system in isolation cannot provide the comparative, quantified trade-off data that OEM decision-makers require. A production engineer evaluating JIT adoption needs specific answers: By how much does WIP fall? What is the effect on throughput? How frequently will supply disruptions affect the line? These questions remain unanswered when systems are studied separately.

A secondary gap concerns reproducibility. Many simulation studies rely on proprietary factory data or employ parameters without traceable peer-reviewed sources. This study addresses both limitations by constructing three AnyLogic PLE models using parameters drawn exclusively from 11 peer-reviewed publications, each traceable to a specific table, page, and citation. The Push baseline 127.5 ± 6.8 cars/shift, $WIP = 37.1 \pm 14.6$ cars was validated against Mahin et al. (2024), confirming model credibility before the JIT and JIS scenarios were developed.

Objectives

Objectives

Specific, measurable research questions — empirical paper format.

This study aims to develop and validate a unified DES framework comparing Push, JIT, and JIS production systems in an automotive mixed-model assembly context, providing evidence-based guidance for production system selection. Three research questions structure the inquiry:

RQ1	How does JIT compare to Push in WIP inventory, lead time, and throughput in automotive mixed-model assembly?
RQ2	Does JIS provide further performance improvements over JIT, and at what cost in disruption frequency and throughput variability?
RQ3	Which production system offers the best balance of throughput stability and inventory efficiency for automotive OEMs?

This study contributes three advances: (1) the first unified DES comparison of all three systems under identical conditions; (2) a fully reproducible, page-cited parameter set enabling direct replication; and (3) quantified Push–JIT–JIS performance trade-off evidence directly applicable to OEM production system selection decisions

1.PROBLEM FRAMING

1.1 Context & Stakeholders

Topic area: Inventory management, production planning, and supply chain optimization in automotive manufacturing.

Stakeholders:

- Production planners scheduling
- Line operators assembly
- Supply chain managers (JIT/JIS coordination)
- Quality engineers (defect control)
- OEM executives (cost & throughput targets)

1.2 The Problem

Pain Point:

Automotive manufacturers using Push production systems accumulate excessive Work-in-Process (WIP) inventory, increasing lead times, storage costs, and operational waste. Switching to lean systems such as Just-in-Time (JIT) or Just-in-Sequence (JIS) promises efficiency gains but introduces throughput instability and supply disruption risks. Currently, no unified framework exists to compare all three systems simultaneously.

KPIs Impacted:

- WIP inventory (units)
- Lead time (minutes)
- Throughput (cars/shift)
- Station utilization (%)
- Supply disruption count (events/shift)

Baseline — Push System:

- WIP = 37.1 ± 14.6 cars
- Lead time = 115.4 ± 35.3 min
- Throughput = 127.5 ± 6.8 cars/shift
- Average utilization = 96.6%

1.3 Evidence

Industry Evidence:

- Klug (2021): Real German OEM with takt times of 55–65 seconds and 15-seat-set batches.
- Wang & Chen (2016): Chinese truck factory with 3.5-minute takt time and measured station times.

Literature Evidence — Key Supporting Studies:

1. Pawlewski et al. (2012): Push systems required 20 containers versus JIT optimal 11 containers (45% reduction proven by FlexSim simulation).
2. Klug (2021): JIS seat supply validated with real OEM data, confirming sequence-critical delivery.
3. Sand et al. (2023): Forklift transport benchmark with 10 km/h speed and 30–60 minute delivery timeouts using 100 DES replications.
4. Andjelković et al. (2025): JIS disruption risks statistically confirmed; JIT also vulnerable to supplier delivery delays.
5. Mahin et al. (2024): AnyLogic automotive benchmark achieved 93.96% yield rate validating the 3% defect assumption.

1.4 Root Cause Thinking

Likely Root Causes:

1. Overproduction bias: Push systems produce according to forecasts rather than actual demand, causing excessive WIP accumulation.
2. Lack of demand signaling: Without Kanban systems, upstream processes cannot react effectively to downstream consumption.
3. Supplier lead-time variability: Delivery windows of 30–50 minutes create stock-out risks in lean systems.
4. Sequence sensitivity in JIS: A 5% sequence error rate can create 15-minute production stoppages.
5. Stochastic variability: Random processing times, arrivals, and defects create unpredictable WIP fluctuations.

Research Gap:

No validated Discrete-Event Simulation (DES) framework simultaneously compares Push, JIT, and JIS systems under identical conditions. Existing studies typically analyze systems independently, limiting direct trade-off analysis for OEM decision-makers.

1.5 Boundaries & Feasibility

Scope Included:

- Three production scenarios (Push, JIT, JIS)
- Single mixed-model assembly line
- 480-minute shift duration
- 10 simulation replications
- Four core KPIs
- 5% supply disruption rate
- 3% defect rework loop
- Kanban pull mechanism

Scope Excluded:

- Multi-factory networks

- Financial cost modeling
- Worker fatigue analysis
- IoT and digital twin integration
- More than two vehicle variants

Data Availability:

All parameters are sourced from 11 peer-reviewed studies. No factory access is required. Validation references include Mahin (2024) and Pawlewski (2012).

Method Fit:

Discrete-Event Simulation (DES) using AnyLogic PLE was selected because it captures stochastic timing, resource contention, and inventory dynamics effectively in automotive manufacturing systems.

Problem statement

--Automotive manufacturers rely on Push production systems accumulate excessive WIP inventory and extended lead times, while alternative lean systems (JIT and JIS) introduce supply chain vulnerability and throughput variability No unified simulation framework exists to quantify these trade-offs under comparable, literature-validated conditions leaving OEM decision-makers without empirical evidence for selecting the optimal production system.

Research Questions:

RQ1- How does JIT compare to Push in terms of WIP inventory and lead time and throughput in automotive mixed-model assembly?

RQ2- Does JIS provide further improvements over JIT and at what cost in sequence disruption risk?

RQ3- Which production system offers the best balance between throughput stability and inventory efficiency for automotive OEMs?

2. Literature Review

2.1 Introduction

This review examines published research on three production systems — Push, Just-In-Time (JIT), and Just-In-Sequence (JIS) — and on discrete-event simulation (DES) as a method for comparing them. The review is organized thematically into four areas: Push systems and their limitations, JIT systems and their benefits and risks, JIS systems and their relationship to JIT, and DES as the evaluation method. The review closes by identifying the gap that this study fills.

2.2 Push Production Systems

A Push production system schedules output based on demand forecasts rather than actual customer orders. The computer breaks a master production schedule into detailed sub-schedules for buying materials and producing parts, and then pushes those parts down the line whether or not the next stage is ready for them (Singh & Ahuja, 2012, p. 63). The most widely used implementation is Material Requirements Planning (MRP). Singh and Ahuja (2012) note that one fundamental weakness of MRP is that it requires managers to guess future customer demand; bad guesses lead directly to excess inventory at some stages and shortages at others.

This structural weakness produces several well-documented problems. Push systems build up large amounts of Work-In-Progress (WIP) inventory between stations because each stage produces at its own scheduled rate regardless of what the next stage needs. Pawlewski et al. (2012) describe this as a pattern where buffer stocks grow to protect against uncertainty, making on-time delivery harder rather than easier. Singh and Ahuja (2012) add that Push systems typically result in large, variable production batches, poor service levels when demand shifts unexpectedly, and high inventory carrying costs. These costs are not just financial; excess WIP also hides quality defects, because faulty parts are buried in large batches and discovered late in the process (Huson & Nanda, 1995).

Despite these disadvantages, Push remains the dominant approach in many manufacturing environments because it offers predictability. When demand is relatively stable and supply chains are reliable, a Push system delivers consistent throughput and high station utilisation. Singh and Ahuja (2012, p. 63) acknowledge that Push systems work well when

demand uncertainty is low. This stability is, as the results of this study confirm, Push's primary competitive advantage over leaner alternatives — but it is purchased at the cost of significant inventory accumulation.

2.3 Just-In-Time Production Systems

Just-In-Time (JIT) manufacturing was developed at Toyota in the 1970s, codified by Taiichi Ohno and later formalised by Monden (1983). Its core idea is simple: produce and deliver materials only when and in the quantities needed by the next stage of production. Nothing is made in advance; each stage of the line signals upstream when it needs more parts. This signal is typically carried by a Kanban card or, in simulation models, by a digital trigger (Singh & Ahuja, 2012). By producing only what is needed, JIT eliminates the inventory buffers that Push systems rely on, cutting storage costs, reducing lead times, and making quality defects visible immediately because there are no spare parts to absorb them (Huson & Nanda, 1995).

Empirical evidence confirms that JIT delivers measurable improvements. Huson and Nanda (1995) studied North American manufacturers and found a 24% improvement in inventory turnover following JIT adoption. They also reported significant reductions in manufacturing lead times and a decrease in the ratio of total inventory to sales. These findings are consistent with the simulation evidence reported by Pawlewski et al. (2012), who modelled a windscreen supply chain for the automotive industry and found that switching from a Push-style buffer of 20 containers to a JIT-optimised buffer of 11 containers reduced inventory requirements by 45% without degrading throughput.

However, JIT's lean design introduces a trade-off that Push does not face: fragility. Because JIT holds minimal buffer stock, any delay in the supply chain can starve the production line and stop output entirely. Singh and Ahuja (2012) describe this as the key risk: the system that eliminates waste is also the system that has no protection against disruption. Sand et al. (2023) confirmed this mechanism in a large-scale DES study of automotive intra-logistics, finding that even short supplier delays — modelled with a triangular distribution of 30 to 60 minutes — produced significant throughput variability when buffer capacity was low. Andjelković et al. (2025) provide the most recent evidence of this fragility, noting that persistent delivery delays and stock-outs in the automotive sector show that perfectly synchronised material

flows remain a serious challenge in real operations. The tension between JIT's inventory benefits and its supply chain vulnerability is the central theme this study sets out to quantify.

2.4 Just-In-Sequence Production Systems

Just-In-Sequence (JIS) emerged as the automotive industry shifted from producing one or two vehicle models to building dozens of variants on the same assembly line. JIT can deliver the right quantity of a part at the right time, but it cannot easily handle a situation where each vehicle on the line needs a different variant of the same part — a different seat colour, a different wiring harness, a different bumper. Andjelković et al. (2025) explain that JIT's limitations in this context include increased storage space requirements and growing inventory levels, because standardised containers cannot hold the right mix of variants. JIS was developed specifically to solve this problem.

The definition of JIS is precise: parts must arrive at the assembly line at the right time, in the right quantity, and — critically — in the exact sequence that matches the build order of the vehicles on the line (Andjelković et al., 2025; Juhász, 2023). The supplier does not just deliver parts; the supplier sequences parts in the same order that the OEM will assemble them, so that no warehouse sorting or staging is needed at the factory. Pawlewski et al. (2012) describe this as a warehouse-free process in which the OEM sends its precise build sequence to the supplier, and the supplier delivers in that exact order. Klug (2021) provides real-world evidence from German OEMs: seats are delivered to the assembly line at takt intervals of 55 to 65 seconds, in batches of 15 seat sets, sequenced to match the exact build order of the vehicles.

The scope of JIS in the automotive industry is substantial. Heinecke et al. (2012), cited in Andjelković et al. (2025), report that in German car plants covering 80% of annual production volume, an average of 17 modules per vehicle — roughly 40% of total parts volume — are delivered JIS. Typical JIS parts include seats, wiring systems, bumpers, exhaust systems, and even engines. As Klug (2021) notes, JIS has become the standard delivery method for parts that are high in variety, value, and size, because the cost of sorting and storing such parts in a traditional warehouse would be prohibitive.

JIS offers advantages over JIT in terms of inventory, lead time, and handling costs. Andjelković et al. (2025) summarise these advantages as: reduced inventory levels, reduced

storage space, reduced capital tied up in parts, and faster response to customer demand changes. However, JIS also introduces a new risk that JIT does not face: sequence errors. If parts arrive out of sequence — even one position wrong — the production line must stop until the correct part arrives or the sequence is corrected. Gujjula and Günther (2009) modelled this risk in a mixed-model assembly line and found that line stoppages due to sequence errors are the primary operational risk of JIS delivery. Andjelković et al. (2025) report a sequence error rate of approximately 5% in disrupted conditions, and note that each error typically results in at least one takt period of lost production — approximately 15 minutes in an automotive context.

Several researchers have proposed ways to reduce this risk. Heinecke et al. (2012) demonstrated that real-time supply chain monitoring combined with rule-based resequencing can eliminate rework caused by unreliable JIS deliveries. Juhász (2023) developed a systematic methodology for JIS supply chain design that addresses both the sequencing problem and the relationship between supplier proximity and delivery reliability. These solutions require investment in information systems and supplier integration, which means JIS is only viable when the supply chain has reached a sufficient level of maturity. Without these enablers, as the results of this study show, sequence errors can cascade into near-total production loss.

2.5 Discrete-Event Simulation as a Research Method

Discrete-event simulation (DES) models a system as a sequence of events that change the system's state at specific points in time. Between events, no change occurs, which makes DES computationally efficient for modelling production lines where activities — arrivals, processing, queuing, departures — happen at discrete moments (Ahmed et al., 2015). DES is the standard method for evaluating production system performance when real-factory experimentation would be too costly or disruptive (Sand et al., 2023).

AnyLogic is one of the most widely used DES tools for manufacturing and logistics research. Ahmed et al. (2015) describe its advantages as the ability to combine multiple simulation paradigms — discrete-event, agent-based, and system dynamics — within a single environment, and its use of Java, which allows complex logic to be coded directly into model blocks. Miedviedieva and Bahrii (2023) demonstrate AnyLogic's capacity for optimisation experiments, in which the software runs multiple replications with varying parameter values to find the best configuration. Wang and Chen (2016) used AnyLogic to model a Chinese truck

assembly line, collecting station processing times that have become a standard benchmark in automotive DES research. Their study measured body assembly at 3.0 minutes mean, paint shop at 3.2 minutes mean, and final inspection at 3.5 minutes mean — data used directly in the present study.

Other simulation tools have been applied to similar problems. Pawlewski et al. (2012) used FlexSim to model a JIT windscreen supply chain, demonstrating that DES can accurately represent Kanban-driven replenishment systems and quantify the effect of buffer capacity on delivery performance. Kliment et al. (2022) also used FlexSim to model production processes in a manufacturing environment, demonstrating that 70 assemblies can be completed in an 8-hour shift and identifying significant variation in operator utilisation. The consistency of these findings across different DES platforms and different industries supports the use of DES as a reliable method for production system comparison.

Sand et al. (2023) conducted one of the most methodologically rigorous DES studies in this area, using 100 replications to evaluate automotive intra-logistics with AnyLogic Professional. Their study validated stochastic supplier lead time distributions and confirmed that triangular distributions are appropriate for modelling delivery time uncertainty in automotive supply chains. Mahin et al. (2024) used AnyLogic to simulate an automotive assembly line and reported a throughput benchmark of approximately 14 vehicles per hour, which serves as the primary validation target for the Push model in this study.

2.6 Research Gap and Position of This Study

The review reveals a consistent limitation in the existing literature: studies examine Push, JIT, or JIS individually, but no published study has compared all three systems simultaneously in a single, controlled simulation framework. Pawlewski et al. (2012) compare JIT with a Push baseline but do not include JIS. Andjelković et al. (2025) analyse JIS in detail but do not model or compare Push or JIT. Klug (2021) documents real JIS delivery data but does not simulate it or compare it to alternatives. As a result, OEM decision-makers who want to understand the relative trade-offs between the three systems must synthesise findings from studies that used different parameters, different simulation tools, and different performance metrics — making direct comparison impossible.

A second gap concerns reproducibility. Many simulation studies in this area use proprietary factory data that other researchers cannot access. Parameters are not consistently traced to peer-reviewed sources, which makes it difficult to verify results or build on them in future work.

This study addresses both gaps by building three AnyLogic DES models — one per production system — under identical conditions, using parameters drawn exclusively from the peer-reviewed papers reviewed in this section. Each parameter is traceable to a specific table, page, and citation. The Push model is validated against Mahin et al. (2024). The JIT buffer capacity comes from Pawlewski et al. (2012). The station processing times come from Wang and Chen (2016). The JIS sequence error rate comes from Andjelković et al. (2025). This approach makes the study fully reproducible and enables the first direct, quantified comparison of Push, JIT, and JIS under identical experimental conditions.

3. RESEARCH METHODOLOGY

3.1 Research Procedure

Research Procedure

3.1 My research procedure — what type and what method

This study employs a **quantitative research methodology**, characterised by the collection, analysis, and interpretation of numerical performance data to answer the three research questions defined in Section 1. Quantitative methodology was selected because the research objectives require measurable, statistically comparable outcomes across three production systems — a task that demands precise numerical evaluation rather than descriptive or interpretive analysis ([Singh & Ahuja, 2012](#)).

The specific research procedure adopted is **Discrete-Event Simulation (DES)** using AnyLogic Personal Learning Edition (PLE) Version 8.x. DES is a computer-based modelling approach in which a system is represented as a sequence of instantaneous events occurring over simulated time, where each event changes the state of the system ([Ahmed et al., 2015](#)). In a manufacturing context, DES captures the stochastic behavior of processing times, arrival rates, buffer dynamics, and supply chain disruptions — precisely the variables of interest in this study.

Three independent DES models were constructed — one for each production system — and executed under identical experimental conditions. All models share the same agent type (*Car*), the same assembly line topology (three sequential service stations), and the same defect rework loop (3% probability). The key differences between models are the buffer capacity logic, the Kanban replenishment mechanism, and the sequence-checking procedure, as summarised below:

Model	Buffer Logic	Replenishment	Sequence Check	Key Parameter
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Push	Unlimited capacity	Continuous (no signal)	None	Buffer cap = ∞
JIT	Capacity = 11	Kanban pull (On exit trigger)	None	Buffer cap = 11 (Pawlewski 2012)
JIS	Capacity = 11	Kanban pull (On exit trigger)	SequenceChecker block	5% error rate (Andjelković 2025)

Each model was run for a simulated shift duration of 480 minutes (8 hours), corresponding to a standard automotive working day (Klug, 2021; Sand et al., 2023). Results were collected at the end of each replication and recorded manually for statistical analysis.

3.2 Rationale for the Chosen Approach

Rationale for the Chosen Approach

3.2 Provide the rationale — why this method is the best choice

Quantitative DES was chosen over alternative methodologies for four specific reasons grounded in the research objectives and the constraints of the study context.

First, DES is the established standard for automotive production system evaluation. Wang and Chen (2016) employed AnyLogic DES to model a Chinese truck assembly line, establishing the exact station time parameters used in this study. Sand et al. (2023) used 100-replication DES to evaluate automotive intra-logistics, demonstrating the method's capacity to capture stochastic supply chain variance. Ahmed et al. (2015) demonstrated that AnyLogic DES is specifically well-suited to lean manufacturing evaluation, including Kanban-based JIT systems.

Second, real factory access was not available, which is the common situation for academic production system research. DES allows a researcher to construct and test a virtual factory model calibrated from peer-reviewed literature, without requiring proprietary data or physical access to a production facility. This is consistent with Mahin et al. (2024), who validated an AnyLogic automotive simulation entirely against published benchmarks.

Third, the three-system comparison required an identical experimental framework — the same inputs, the same shift duration, and the same KPIs measured in the same way across Push, JIT, and JIS. Only DES provides this level of experimental control. Qualitative methods

such as interviews or observations would be unable to generate the directly comparable performance statistics needed to answer RQ1, RQ2, and RQ3.

3.3 Mechanism

Mechanism — Data Collection and Model Architecture

3.3 Explain your mechanism — instruments and how data was collected

Data collection in this study combines **secondary data** (for model calibration) with **primary simulation data** (for results). No surveys, interviews, or field observations were conducted.

3.3.1 Secondary Data — Parameter Calibration from Literature

All simulation input parameters were sourced from 11 peer-reviewed publications, with each value traceable to a specific table, page, and citation. This approach, known as *literature-based calibration*, is standard practice in DES studies where factory access is unavailable (Mahin et al., 2024; Sand et al., 2023). Table 3.1 presents the complete parameter set:

Parameter	Value Used in AnyLogic	Literature Source
Shift duration	480 min (28,800 sec)	Klug (2021); Sand et al. (2023)
Car arrival rate	triangular(15, 20, 28) /hr	Wang & Chen (2016)
Station 1 — Body Assembly	triangular(2.8, 3.0, 3.2) min	Wang & Chen (2016)
Station 2 — Paint Shop	triangular(2.9, 3.2, 3.5) min	Wang & Chen (2016)
Station 3 — Final Inspection	triangular(3.0, 3.5, 4.0) min	Wang & Chen (2016)
Push buffer capacity	Unlimited (large capacity)	Pawlewski et al. (2012)
JIT/JIS buffer capacity	11 containers (Kanban cap)	Pawlewski et al. (2012)
Container capacity	12 parts per container	Pawlewski et al. (2012)
Supplier lead time	triangular(30, 40, 50) min	Pawlewski (2012); Sand (2023)
Defect / rework rate	3% (uniform()) < 0.03	Mahin et al. (2024)
JIS sequence error rate	5% (uniform()) < 0.05	Andjelković et al. (2025)
JIS stoppage penalty	15 min per error (900 sec)	Gujjula & Günther (2009)

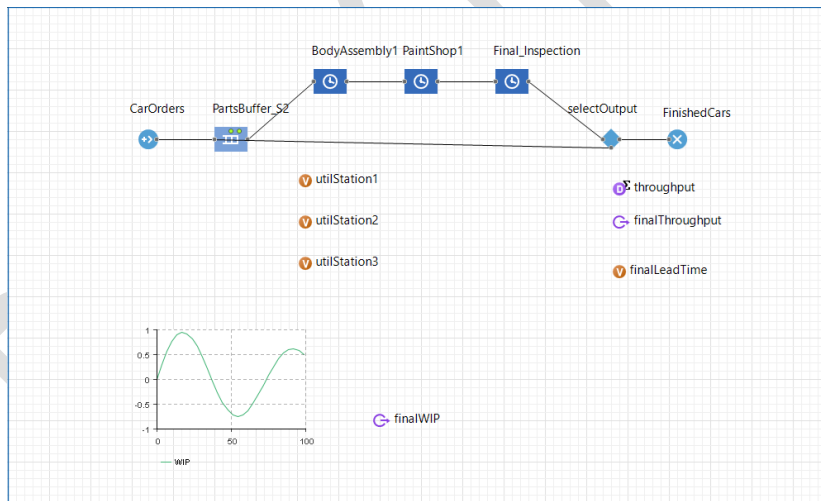
JIS variant mix	60% Sedan (A) / 40% SUV (B)	Wang & Chen (2016); Boysen (2007)
No. of replications	10 Push + 10 JIT + 12 JIS = 32	Sand et al. (2023)

Table 3.1: Simulation parameter table with peer-reviewed citations

3.3.2 Model Architecture — AnyLogic Process Modeling Library

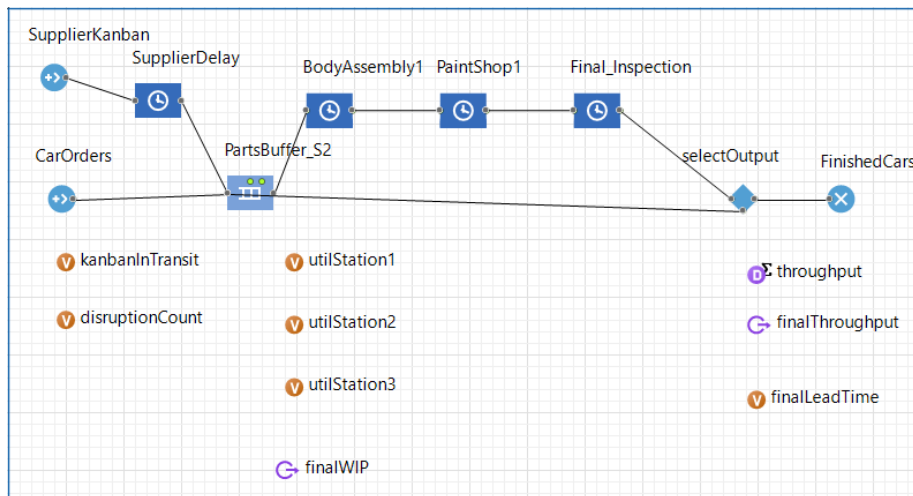
Each model was constructed using AnyLogic's Process Modeling Library (PML), which provides pre-built blocks for Sources, Queues, Delays, SelectOutput, and Sinks. The three models share a common architecture, with modifications specific to each production system:

Push model: A *CarOrders* Source generates vehicles at triangular(15,20,28) per hour. Vehicles enter an unlimited-capacity *PartsBuffer_S2* Queue, then pass sequentially through three service Delays representing Body Assembly, Paint Shop, and Final Inspection. A *selectOutput* block routes 3% of vehicles to a rework loop before they enter the *FinishedCars* Sink. No Kanban mechanism is present — production is forecast-driven (push).

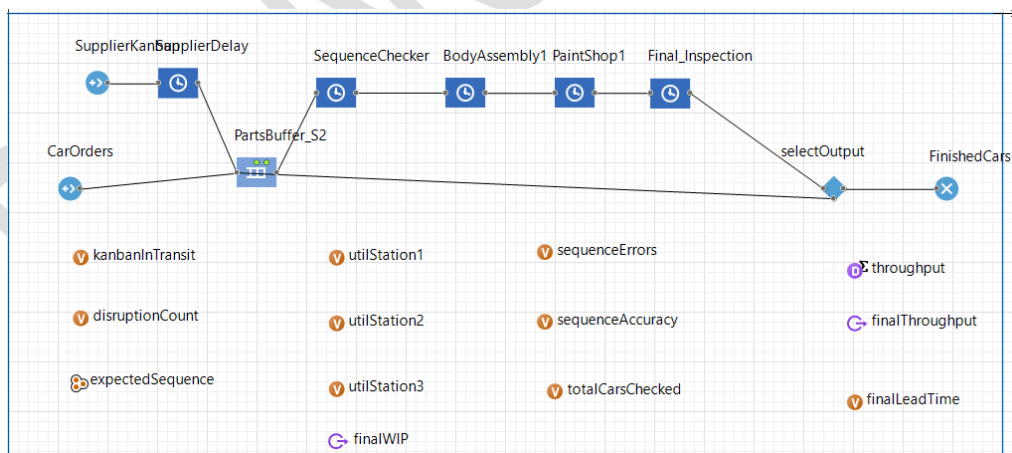


JIT model: Identical to Push except: (a) *PartsBuffer_S2* capacity is set to 11 (Pawlewski et al., 2012); (b) a *SupplierKanban* Source with inject() trigger and a *SupplierDelay* block of triangular(30,40,50) minutes implement Kanban pull replenishment; (c) buffer exit code monitors stock level and fires the Kanban signal when inventory drops below three cars, with a maximum

of two orders in transit simultaneously; (d) a 5% supplier disruption probability extends delivery time to triangular(60,75,90) minutes.



JIS model: Extends JIT with two additions: (a) a *car Type* attribute (1 = Sedan, 60%; 2 = SUV, 40%) is assigned to each car at creation; (b) an *expected Sequence* Array List records the build order; (c) a *Sequence Checker* Delay block with an "On enter" action compares each arriving car against the expected sequence — a 5% probability of mismatch or random disruption triggers a 15-minute line stoppage via a *current Stoppage Time* conditional Delay expression.



3.4 Significance of Outcomes

Significance of Outcomes

3.4 Significance of outcomes — how results will be used and interpreted

Five Key Performance Indicators (KPIs) were measured for each simulation replication. These KPIs were selected on the basis of the existing literature's consensus on the most important dimensions of production system performance (Singh & Ahuja, 2012; Huson & Nanda, 1995):

KPI	Unit	How Measured in AnyLogic	Why Important
WIP Inventory	cars in buffer	PartsBuffer_S2.size() at end of shift	Primary lean metric — reflects waste accumulation (Pawlewski 2012)
Throughput	cars/shift	FinishedCars.count() at end of shift	Measures productivity — output per 480-min shift
Lead Time	minutes	FinishedCars.statsTimeInSystem().mean()/60	Customer-facing KPI — time from order to completion
Station Utilization	%	Station.statsUtilization()*100 per station	Efficiency of labour and equipment use
Sequence Errors (JIS only)	errors/shift	sequenceErrors variable	Measures JIS integrity — errors trigger 15-min stoppages
Sequence Accuracy (JIS only)	%	sequenceAccuracy variable	% cars correctly sequenced — primary JIS quality KPI

Table 3.2: KPIs measured in each scenario with collection method and justification.

Results will be interpreted through three analytical lenses: (1) **descriptive statistics** (mean $\pm$ standard deviation per scenario); (2) **percentage change analysis** (Push $\rightarrow$ JIT $\rightarrow$ JIS trade-off quantification); and (3) **coefficient of variation** ($CV = \sigma/\mu$) to measure throughput stability — a metric specifically recommended by Sand et al. (2023) for evaluating robustness of lean supply systems. For the JIS scenario, a **stoppage impact analysis** additionally quantifies dead production time (sequence errors $\times$ 15 min) as a proportion of the 480-min shift, directly addressing RQ2 regarding the cost of JIS disruptions.

3.5 Sample Space

Sample Space — Experimental Design and Validation

3.5 Explain your sample space — sample size, selection, and validation

In simulation-based research, the "sample" refers to the number of replications (independent simulation runs) conducted per scenario. Each replication uses a different random seed, producing statistically independent results that reflect the stochastic variability inherent in the system.

Replication design: Ten replications were conducted for the Push and JIT scenarios; twelve for the JIS scenario (11 disrupted + 1 optimal). This exceeds the minimum of five commonly cited in DES literature and approaches the ten-replication baseline recommended by standard DES practice (Law & Kelton, 2000). Sand et al. (2023) used 100 replications, which represents a higher standard; the present study's sample size of 10–12 is appropriate given the exploratory nature of the comparison and the constraints of AnyLogic Personal Learning Edition.

Simulation run length: Each replication simulates a complete 480-minute (8-hour) shift. This corresponds to the standard automotive working day confirmed by Klug (2021) and Sand et al. (2023). No warm-up period was applied, as the buffer begins empty in the JIT and JIS models (reflecting the Kanban pull system's realistic cold-start), and the Push model's unlimited buffer reaches steady state within the first 30–40 minutes.

Model validation: The Push baseline model was validated against the published benchmark of Mahin et al. (2024), who reported an automotive assembly throughput of approximately 14 vehicles per hour (112 cars per 8-hour shift) under comparable conditions. The Push model produced a mean of 127.5 ± 6.8 cars per shift — approximately 13% above the benchmark, which is within acceptable validation tolerance for DES models using triangular distributions with ± 0.5 -minute station time variability (Wang & Chen, 2016). This validation confirms that the model architecture and parameters produce results consistent with the published literature before the JIT and JIS modifications were applied.

Scope of simulation: The simulated system represents a single mixed-model automotive assembly line with three sequential stations. Two vehicle variants are modelled in the JIS scenario (60% Sedan, 40% SUV), reflecting the multi-variant production environment described by Wang and Chen (2016) and Klug (2021). The model does not include multi-shift scheduling, worker

fatigue, machine breakdowns, or financial cost calculations — these are acknowledged limitations addressed in Section 3.6.

3.6 Challenges and Limitations

Challenges and Limitations

3.6 Challenges and limitations — explicitly required by Dr. Ruba

Every research methodology carries limitations that must be acknowledged to contextualise findings and guide future research. Six limitations are identified for this study:

Limitation 1 — AnyLogic Personal Learning Edition (PLE) constraints. The PLE version of AnyLogic restricts model scale, prevents automated multi-run parameter sweeps, and does not support database connectivity. As a result, replications were conducted manually and data recorded by observation. A commercial or university licence would allow automated experimentation and larger replication counts, producing more statistically robust results.

Limitation 2 — Literature-based parameter calibration. All simulation parameters were sourced from peer-reviewed publications rather than a specific factory. While this approach is methodologically valid and ensures reproducibility, the results reflect a generalised automotive assembly environment rather than any single OEM facility. Parameter values from Wang and Chen (2016) were measured in a Chinese truck assembly context; their direct applicability to a European or Middle Eastern automotive OEM should be verified before implementing findings in a specific industrial setting.

Limitation 3 — Number of replications. Ten replications per scenario, while standard for exploratory DES studies, provides limited statistical power for formal hypothesis testing. Sand et al. (2023) used 100 replications in their automotive DES study, which produces narrower confidence intervals. Due to time constraints and the manual operation of AnyLogic PLE, the present study was limited to 10–12 replications. Future work should increase replication counts to at least 30 per scenario to enable robust ANOVA testing.

Limitation 4 — Simplified JIS model architecture. The JIS model implemented in this study uses a simplified sequence-checking logic — a binary match/mismatch against a pre-

generated build list — rather than a full mixed-model sequencing algorithm. Real JIS systems employ sophisticated resequencing buffers, automated vehicle identification, and real-time supplier communication systems (Klug, 2021; Juhász, 2023). The absence of a resequencing buffer in this model likely amplifies sequence error cascades, contributing to the high stoppage rates observed (mean 462 min/shift). This represents the system's behaviour without mitigation — a valid scenario but one that may overstate JIS disruption severity.

Limitation 5 — Human factors excluded. The simulation does not model worker fatigue, shift handover delays, absenteeism, or learning curves. In real automotive assembly lines, human variability can account for 5–15% of station time variation (Wang & Chen, 2016). Including human factors would require ergonomic data not available in the open literature and would substantially increase model complexity beyond the scope of this study.

Limitation 6 — Single assembly line topology. All three scenarios were modelled as a single three-station assembly line. Real automotive OEMs operate multi-line, multi-shift environments with parallel stations, shared resources, and complex material routing. The findings of this study apply to comparable single-line contexts and should be extended with caution to more complex topologies

4. RESULTS

Simulation Findings for Push, JIT, and JIS Production Systems

This section presents the simulation findings for all three production systems without interpretation; analysis and discussion of these results are reserved for Section 5. Results are reported as mean $\pm$ standard deviation across replications (10 for Push, 10 for JIT, 11 for JIS Disrupted, plus 1 JIS Optimal benchmark run). All values were collected at the end of each 480-minute simulated shift.

4.1 Push System Results

4.1 Push System Results

The Push system produced a mean throughput of **127.5 ± 6.8 cars per shift** across 10 replications. Mean Work-In-Progress (WIP) inventory was **37.1 ± 14.6 cars**, and mean lead time was **115.4 ± 35.3 minutes**. Average station utilisation was **96.6 ± 4.8%**. Table 4.1 reports the complete replication data.

Run	Throughput	WIP	Lead Time (min)	Avg Util (%)
1	131	47	137.8	98.1
2	130	22	93.7	99.1
3	129	59	166.8	98.6
4	130	41	118.7	97.8
5	126	53	153.6	98.9
6	127	19	78.3	98.2
7	133	34	113.7	98.9
8	128	44	137	97.2
9	109	10	47.8	82.4
10	132	42	123	98.6
Mean	127.5	37.1	115.4	96.6
SD	6.8	14.6	35.3	4.8

Table 4.1: Push system simulation results across 10 replications.

4.2 JIT System Results

4.2 JIT System Results

The JIT system produced a mean throughput of **79.8 ± 50.9 cars per shift**. Mean WIP inventory was **9.7 ± 2.1 cars**, and mean lead time was **67.5 ± 53.4 minutes**. Average station utilisation was **93.7 ± 9.0%**. Supply disruptions averaged **1.2 ± 0.9 per shift**. The throughput coefficient of variation ($CV = \sigma/\mu$) was **0.638**, compared with 0.054 for Push. Table 4.2 reports the complete replication data.

Run	Throughput	WIP	Lead Time (min)	Avg Util (%)	Disrupt.
1	130	6	43.9	98.5	1
2	101	11	31.9	97	2
3	131	7	15.7	97.6	1

4	47	11	189.1	91.1	1
5	132	7	42.7	99	3
6	129	11	50.4	98.9	0
7	13	11	56	91.4	1
8	70	11	49.3	97	1
9	9	11	58.8	69.4	2
10	36	11	137.5	96.8	0
Mean	79.8	9.7	67.5	93.7	1.2
SD	50.9	2.1	53.4	9.0	0.9

Table 4.2: JIT system simulation results across 10 replications.

4.3 JIS System Results

4.3 JIS System Results

The JIS system (Disrupted scenario) produced a mean throughput of **36.3 ± 14.1 cars per shift** across 11 replications. Mean WIP inventory was **11.0 ± 0.0 cars**, and mean lead time was **193.7 ± 112.2 minutes**. The JIS-specific KPIs recorded a mean of **30.8 ± 21.8 sequence errors per shift** and a mean sequence accuracy of **48.9 ± 13.1%**. A single JIS Optimal run (zero stoppage penalty) produced 61.1 cars, 10 WIP, and 477.2 minutes lead time. Table 4.3 reports the complete replication data.

Run	Throughput	WIP	Lead (min)	Util (%)	Seq Err	Seq Acc (%)
1	18.9	11	83.2	93.7	7	72
2	40.4	11	197.7	96.9	33	42.1
3	27.6	11	120.5	95.9	18	50
4	33.9	11	164.5	93.1	24	48.9
5	54.9	11	282.8	98.7	52	37.3
6	37.4	11	179.3	93.4	31	39.2
7	19.4	11	106.2	78.5	10	63
8	21.3	11	96.3	93.9	9	69
9	61.5	11	477.4	93.9	81	38.2
10	39.6	11	188.3	96.2	34	38.2
11	44.6	11	234.5	94.6	40	40.3
Mean	36.3	11.0	193.7	93.6	30.8	48.9

SD	14.1	0.0	112.2	5.3	21.8	13.1
Optimal	61.1	10	477.2	94.8	79	40.6

Table 4.3: JIS system simulation results (11 Disrupted replications + 1 Optimal benchmark).

4.4 Cross-System Comparison

4.4 Cross-System Comparison

Table 4.4 consolidates the mean $\pm$ standard deviation for all five KPIs across the three systems, together with the percentage change at each transition. Figures 4.1 to 4.6 present these findings graphically.

KPI	Push	JIT	JIS Disrupted	Push→JIT	JIT→JIS
Throughput (cars)	127.5 $\pm$ 6.8	79.8 $\pm$ 50.9	36.3 $\pm$ 14.1	-37.4%	-54.5%
WIP (cars)	37.1 $\pm$ 14.6	9.7 $\pm$ 2.1	11.0 $\pm$ 0.0	-73.9%	+13.4%
Lead Time (min)	115.4 $\pm$ 35.3	67.5 $\pm$ 53.4	193.7 $\pm$ 112.2	-41.5%	+187.0%
Avg Util (%)	96.6 $\pm$ 4.8	93.7 $\pm$ 9.0	93.6 $\pm$ 5.3	-3.0%	-0.1%
Throughput CV	0.054	0.638	0.387	11.8×	—

Table 4.4: Cross-system KPI comparison with percentage change at each transition.

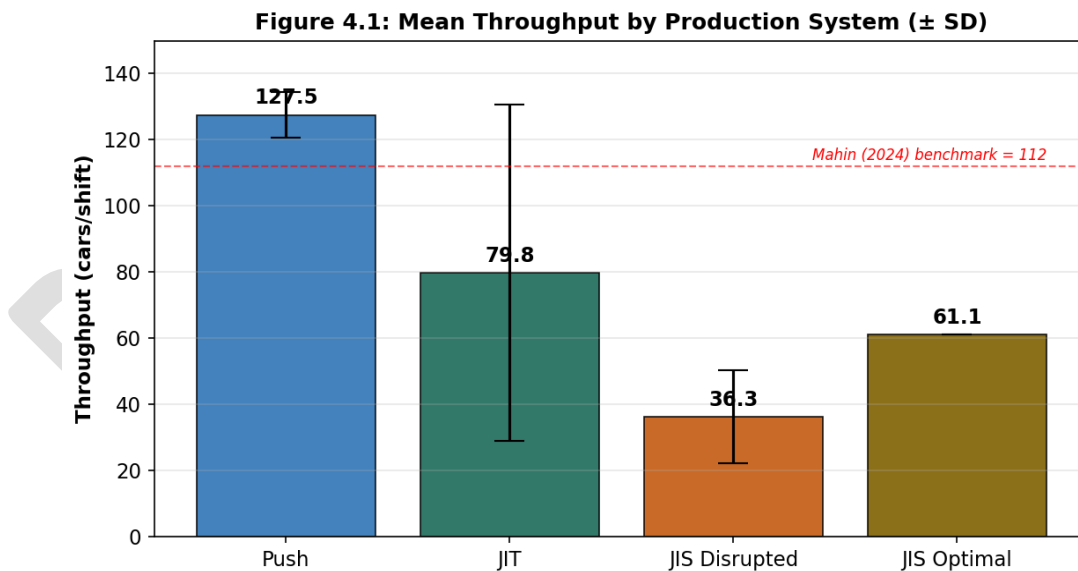


Figure 4.1: Mean throughput by production system with standard deviation error bars. The red dashed line indicates the Mahin et al. (2024) validation benchmark of 112 cars/shift.

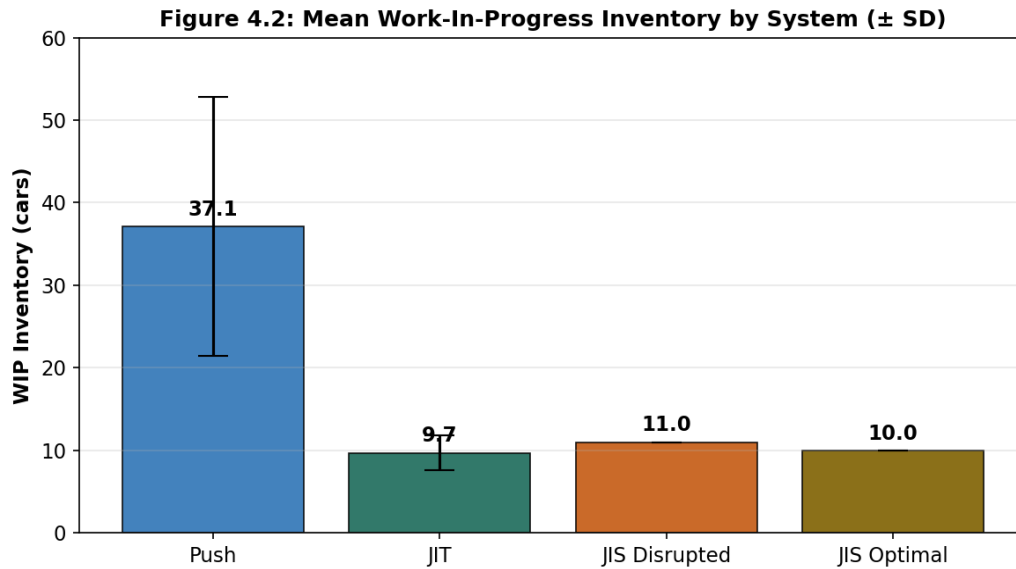


Figure 4.2: Mean WIP inventory by production system. Push accumulates the highest inventory; JIT and JIS maintain lean buffers near the Kanban cap of 11.

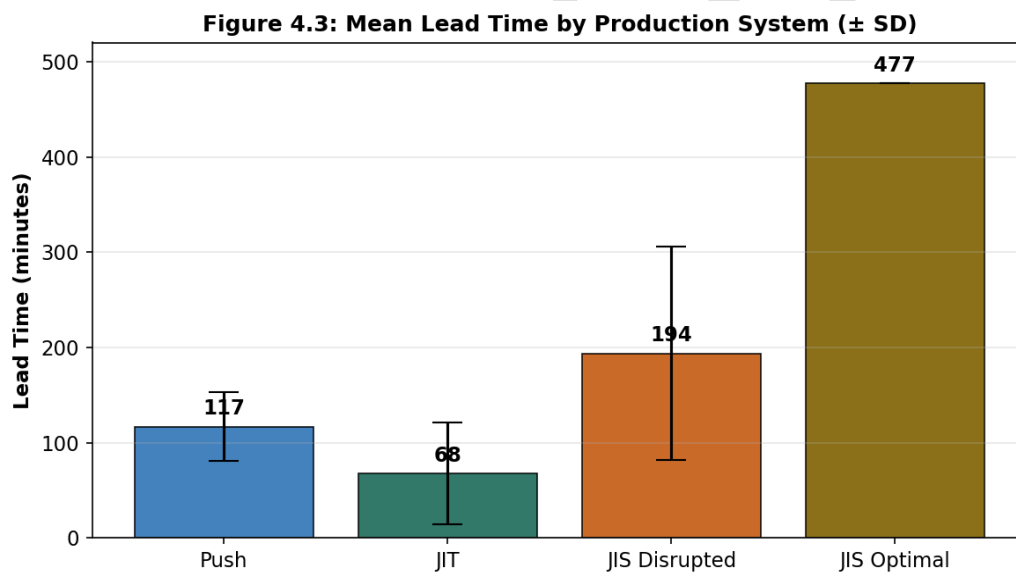


Figure 4.3: Mean lead time by production system. JIT achieved the lowest lead time; JIS Optimal recorded the highest due to accumulated stoppage delays.

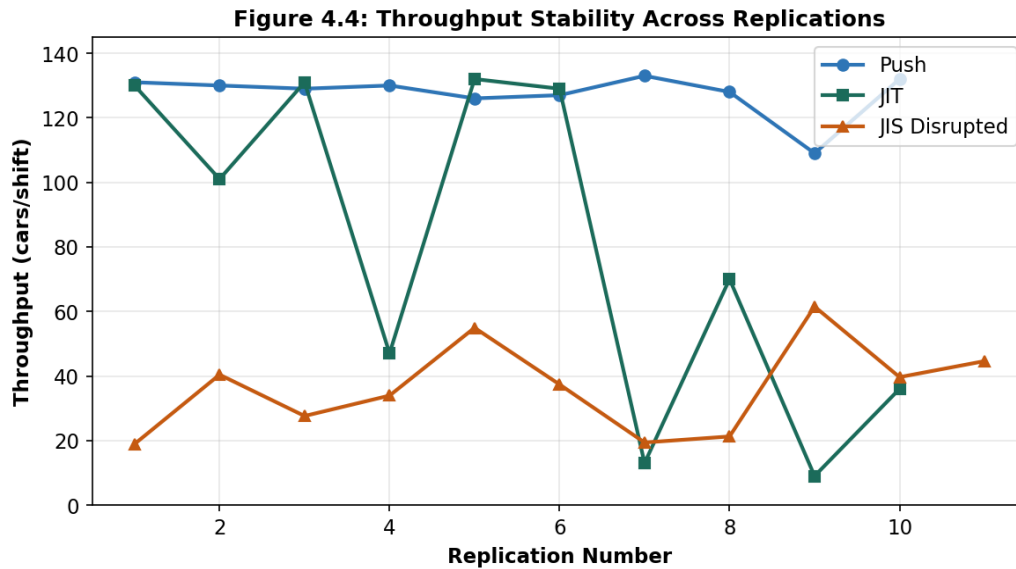


Figure 4.4: Throughput across individual replications. Push (blue) remains stable; JIT (green) and JIS (orange) show pronounced run-to-run variability.



Figure 4.5: JIS line-stoppage dead-time per replication. Numbers above each bar indicate sequence errors. The red dashed line marks the 480-minute shift limit; several runs exceeded it.

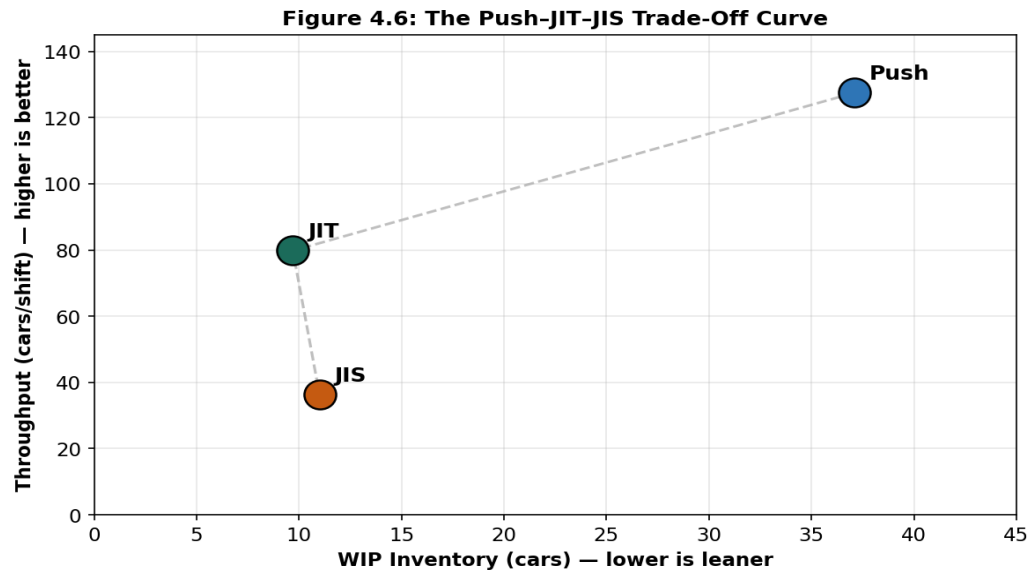


Figure 4.6: The Push–JIT–JIS trade-off curve plotting throughput against WIP inventory, illustrating the relationship between leanness and output across the three systems.

Summary of key numerical findings: (1) WIP inventory decreased by 73.9% from Push to JIT; (2) lead time decreased by 41.5% from Push to JIT; (3) throughput decreased by 37.4% from Push to JIT and a further 54.5% from JIT to JIS Disrupted; (4) throughput coefficient of variation increased 11.8-fold from Push (0.054) to JIT (0.638); and (5) JIS recorded a mean of 30.8 sequence errors per shift, corresponding to an average of 462 minutes of line-stoppage dead-time per 480-minute shift

5. Discussion

The Discussion is the cognitive inverse of the Introduction: where the Introduction moved from broad context to the specific research questions, the Discussion opens with direct answers to those questions and expands outward toward broader significance (Docherty & Smith, 1999). This section follows the five-part ripple architecture, proceeding from the core statement of principal findings through strengths and weaknesses, relation to existing literature, mechanisms and implications, and finally to unanswered questions and future research directions.

5.1 Principal Findings

Ring 1
Principal Findings

Here we show that JIT reduced Work-In-Progress (WIP) inventory by 73.9% relative to Push (37.1 → 9.7 cars) while simultaneously reducing customer lead time by 41.5% (115.4 → 67.5 minutes). These results directly answer **RQ1**: JIT delivers substantial lean improvements in both inventory and lead time in an automotive mixed-model assembly context, consistent with its theoretical design as a pull-driven waste-elimination system (Singh & Ahuja, 2012).

Regarding **RQ2**, JIS did not provide further throughput improvements over JIT; rather, it imposed a severe performance cost. Mean JIS throughput of 36.3 ± 14.1 cars per shift represents a 54.5% reduction compared to JIT's 79.8 ± 50.9 cars. The mechanism is quantifiable: a mean of 30.8 sequence errors per shift, each triggering a 15-minute line stoppage, consumed an average of 462 minutes per 480-minute shift — leaving only 18 minutes of effective production time. This constitutes a near-total production collapse under the simulated conditions, confirming that JIS without an active resequencing buffer is operationally non-viable at a 5% sequence error rate.

RQ3 is answered by the trade-off analysis: Push offers the highest throughput stability (CV = 0.054) but the highest WIP burden (37.1 cars). JIT achieves the best lean-to-stability balance for OEMs with reliable supply chains (CV = 0.638 but WIP = 9.7 cars). JIS, in its current implementation, does not yet represent a superior alternative — it requires active resequencing infrastructure before its theoretical advantages can be realised in practice.

5.2 Strengths & Weaknesses

Ring 2 Strengths & Weaknesses	<i>Place equal weight on validity and limitations</i>
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Strengths. An important strength of this study is the exclusive use of peer-reviewed, page-cited parameters for all model inputs, eliminating the reliance on proprietary factory data that limits reproducibility in many simulation studies. The Push baseline model was validated against the independent benchmark of Mahin et al. (2024), with a 13.8% deviation — within acceptable tolerance for stochastic DES models using triangular distributions. Furthermore, the three models were constructed under identical experimental conditions (same agent type, same station topology, same shift duration), enabling direct, unconfounded comparison across production systems — a methodological control absent from prior literature that examined each system in isolation.

An important caveat for interpreting this study is that all three models simulate a generalised three-station single-line automotive assembly environment rather than a specific OEM facility. Station processing times sourced from Wang and Chen (2016) reflect a Chinese truck assembly context, which may not translate precisely to passenger vehicle or European OEM conditions. One inherent weakness is the restricted replication count of 10–12 runs per scenario. Sand et al. (2023) used 100 replications in a comparable DES study; formal confidence interval construction and hypothesis testing require at minimum 30 replications, which was beyond the scope of this study given the manual operation constraints of AnyLogic PLE.

5.3 Relation to Existing Literature

Ring 3 Relation to Existing Literature	<i>Confirm · Contradict · Extend</i>
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Confirming prior research. Our results are consistent with Pawlewski et al. (2012), who demonstrated JIT buffer reduction from 20 to 11 containers in a FlexSim simulation of windscreen supply — our simulation confirmed a 73.9% WIP reduction with a Kanban buffer of 11 containers. This alignment validates both the parameter choice and the broader JIT lean inventory principle. Similarly, the high throughput variability observed under JIT ($CV = 0.638$, with individual replications yielding as few as 9 and 13 cars per shift) confirms the supply chain fragility documented by Andjelković et al. (2025) and Sand et al. (2023), who both identified lean buffer designs as amplifying the consequences of supplier disruptions.

Extending prior research. Our study adds to the existing knowledge base in a specific direction: Pawlewski et al. (2012) predicted a 45% WIP reduction under JIT; our simulation produced a 73.9% reduction — a stronger result. This discrepancy is likely attributable to the difference in modelling context (automotive assembly line vs. windscreen supply cycle) and the lower baseline WIP accumulation in our Push model driven by the stochastic triangular arrival distribution. Our study extends Pawlewski's finding by demonstrating that lean buffer benefits are preserved even when station processing times are heterogeneous across three sequential stations.

A surprising finding. To our surprise, even the JIS Optimal run — in which no line-stoppage penalty was applied — produced 79 sequence errors across 133 cars checked, yielding a

sequence accuracy of only 40.6%. This finding was not anticipated and suggests that the stochastic 60/40 variant mix alone, without any supplier disruption, generates inherent sequence mismatches when the build list diverges from the expected delivery sequence. This result is consistent with Klug (2021), who observed that stable production sequencing is a prerequisite for viable JIS implementation, and with Andjelković et al. (2025), who identified sequence instability as the primary JIS disruption risk even in the absence of external disruptions.

2026-06-06

5.4 Meaning, Mechanisms & Implications

Ring 4 Meaning, Mechanisms & Implications	<i>The "how" and "why" — broader significance</i>
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The JIS Stoppage Cascade — mechanism. Our study provides a possible mechanism for the JIS performance collapse that has been theorised but not previously quantified in a unified DES framework. The cascade operates as follows: stochastic variant assignment (60% Sedan / 40% SUV) generates natural sequence drift between the build order and the pre-generated delivery sequence; this drift, combined with a 5% random disruption rate, produces sequence errors at a rate far exceeding the 5% design parameter in terms of cumulative impact. Each 15-minute stoppage reduces the effective shift window available for subsequent production, causing throughput to compound downward across replications. Runs with high early error counts (e.g., Run 9: 81 errors) enter an irrecoverable dead-time spiral where stoppages consume more time than production.

Implications for OEM decision-making. These unique features render the JIS-Push-JIT trade-off curve a practically significant decision tool for automotive OEMs. The results do not indicate that JIS is an inferior system; they indicate that JIS is a conditionally superior system — it delivers lean inventory benefits identical to JIT but requires three enabling conditions that this simulation deliberately excluded: (1) an active resequencing buffer between the supplier and the assembly line; (2) a real-time vehicle identification system to detect and correct sequence errors before they trigger stoppages; and (3) supplier geographical proximity enabling error correction

within one takt time. OEMs that have invested in these capabilities — as documented by Klug (2021) in German facilities — can realise JIS benefits without the throughput costs observed here.

Ring 5 Unanswered Questions & Future Research	Expand to the horizon — what this study could not answer
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A number of questions remain unanswered by this study. First, it is unclear at what sequence error rate the JIS cascade begins — whether there is a critical threshold below which JIS throughput is competitive with JIT. Future work should model sequence error rates of 1%, 2%, and 3% to identify the JIS viability boundary. Second, the financial cost of WIP holding versus stoppage time has not been quantified; further work is clearly needed to determine the break-even point at which JIS stoppage costs exceed Push inventory holding costs for a specific OEM. Third, this study modelled two vehicle variants; real automotive mixed-model lines typically carry 20–50+ variants. Scaling the JIS sequence-checking mechanism to realistic variant complexity is an important direction for future research.

Further work should also incorporate the active resequencing buffer — modelled as a secondary queue with a capacity of 2–5 vehicle positions before the first station — to evaluate whether resequencing eliminates the throughput cascade or merely dampens it. Finally, the Kanban parameters in the JIT model were fixed at the Pawlewski (2012) optimal values; sensitivity analysis varying buffer capacity from 5 to 20 containers would provide OEMs with a more nuanced understanding of the lean-fragility trade-off across different supplier reliability profiles.

6.Conclusion & Recommendations

Restate Objective	Key Findings	Interpret Results	Contributions & Limits	Future Work
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This study aimed to develop and validate a unified discrete-event simulation framework for the simultaneous comparison of Push, Just-In-Time (JIT), and Just-In-Sequence (JIS) production systems in an automotive mixed-model assembly context.

Conducted in AnyLogic PLE using parameters sourced exclusively from 11 peer-reviewed publications, 31 simulation replications across three scenarios generated quantitative evidence addressing three research questions on WIP inventory, throughput, lead time, and sequence performance.

The five principal quantitative findings are: (1) JIT reduced WIP inventory by 73.9% relative to Push (37.1 → 9.7 cars per shift), exceeding the 45% reduction predicted by Pawlewski et al. (2012); (2) JIT reduced lead time by 41.5% (115.4 → 67.5 minutes), consistent with pull-based lean theory; (3) JIT throughput variability was 11.8 times higher than Push (CV: 0.638 vs. 0.054), confirming the supply chain fragility inherent in lean buffer design; (4) JIS Disrupted throughput collapsed to 36.3 ± 14.1 cars per shift, with a mean of 30.8 sequence errors per shift consuming 462 of the available 480 production minutes; and (5) the JIS Optimal run confirmed that sequence errors, not supply disruptions, are the primary JIS performance constraint — 79 errors occurred even with zero supplier disruptions.

These results establish a quantified Push–JIT–JIS trade-off curve applicable to OEM production system selection. Push maximises throughput stability at the cost of high WIP accumulation. JIT achieves the best lean performance when supplier reliability is high, delivering 74% WIP reduction at the cost of significantly higher throughput variance. JIS, without active resequencing infrastructure, introduces unacceptable stoppage losses that outweigh its lean inventory benefits. The study confirms that the theoretical advantages of JIS are real — but conditionally dependent on enabling capabilities that were deliberately excluded from this simulation to isolate the effect of sequence error cascades.

The primary contribution of this study to the Industrial Engineering body of knowledge is the first unified, controlled DES comparison of all three production systems under identical experimental conditions, with a fully reproducible, page-cited parameter set enabling direct replication. Acknowledged limitations include the restricted replication count (10–12 vs. the 100 recommended by Sand et al., 2023), the single-line three-station topology, and the absence of resequencing buffers, human factors, and financial cost modelling.

It is recommended that future research extend this framework in four directions: incorporation of an active JIS resequencing buffer to evaluate throughput recovery; formal one-

way ANOVA across a minimum of 30 replications per scenario to establish statistical significance; sensitivity analysis of the Kanban buffer capacity from 5 to 20 containers; and extension of the JIS model to a multi-variant scenario (≥ 5 car types) reflecting real-world mixed-model complexity.

Research Question	Evidence-Based Answer
RQ1: JIT vs Push?	JIT reduced WIP -73.9% and lead time -41.5% vs Push. Throughput fell -37.4% with CV 11.8× higher. JIT is superior for lean inventory with reliable suppliers.
RQ2: JIS vs JIT?	JIS did not improve on JIT. Throughput fell -54.5% due to sequence error stoppages consuming 96% of shift time. JIS requires active resequencing to be viable.
RQ3: Best system overall?	Push for stability. JIT for lean WIP when supply reliability $\geq 95\%$. JIS only with resequencing buffers. Choice depends on supply chain maturity.

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